In [2]:
```
import numpy as np
import pandas as pd

df=pd.read_csv("Book3.csv")

df
```

Out[2]:

	Student_ID	Department	Age	Marks	Passed
0	1	CSE	19.0	85.0	Yes
1	2	ECE	20.0	72.0	Yes
2	3	CSE	18.0	65.0	No
3	4	EEE	19.0	78.0	Yes
4	5	ECE	21.0	NaN	Yes
5	6	CSE	20.0	55.0	No
6	7	EEE	NaN	68.0	No
7	8	CSE	19.0	91.0	Yes
8	9	NaN	20.0	74.0	No
9	10	ECE	18.0	88.0	Yes

In [3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student_ID  10 non-null     int64
 1   Department  9 non-null      str
 2   Age         9 non-null      float64
 3   Marks       9 non-null      float64
 4   Passed      10 non-null     str
dtypes: float64(2), int64(1), str(2)
memory usage: 587.0 bytes
```

In [4]:
```
df.Department.mode()
```

Out[4]:
```
0    CSE
Name: Department, dtype: str
```

In [5]:
```
df.Department.mode()[0]
```

Out[5]:
```
'CSE'
```

In [8]:
```
df['Department'] = df['Department'].fillna(df['Department'].mode()[0])

df['Age'] = df['Age'].fillna(df['Age'].median())

df['Marks'] = df['Marks'].fillna(round(df['Marks'].mean()))
df
```

Out[8]:

	Student_ID	Department	Age	Marks	Passed
0	1	CSE	19.0	85.0	Yes
1	2	ECE	20.0	72.0	Yes
2	3	CSE	18.0	65.0	No
3	4	EEE	19.0	78.0	Yes
4	5	ECE	21.0	75.0	Yes
5	6	CSE	20.0	55.0	No
6	7	EEE	19.0	68.0	No
7	8	CSE	19.0	91.0	Yes
8	9	CSE	20.0	74.0	No
9	10	ECE	18.0	88.0	Yes

In [9]:
```
pd.get_dummies(df.Department)
```

Out[9]:

	CSE	ECE	EEE
0	True	False	False
1	False	True	False
2	True	False	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	False	True	False

In [10]:
```
updated_dataset=pd.concat(
    [pd.get_dummies(df.Department),df.iloc[:,[0,2,3,4]]],
    axis=1
)

updated_dataset
```

Out[10]:

	CSE	ECE	EEE	Student_ID	Age	Marks	Passed
0	True	False	False	1	19.0	85.0	Yes
1	False	True	False	2	20.0	72.0	Yes
2	True	False	False	3	18.0	65.0	No
3	False	False	True	4	19.0	78.0	Yes
4	False	True	False	5	21.0	75.0	Yes
5	True	False	False	6	20.0	55.0	No
6	False	False	True	7	19.0	68.0	No
7	True	False	False	8	19.0	91.0	Yes
8	True	False	False	9	20.0	74.0	No
9	False	True	False	10	18.0	88.0	Yes

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   Student_ID  10 non-null     int64  
 1   Department  10 non-null     str    
 2   Age         10 non-null     float64
 3   Marks       10 non-null     float64
 4   Passed      10 non-null     str    
dtypes: float64(2), int64(1), str(2)
memory usage: 588.0 bytes
```

In [12]:
```
updated_dataset['Passed']=updated_dataset['Passed'].replace(['No','Yes'],[0,1])
updated_dataset
```

Out[12]:

	CSE	ECE	EEE	Student_ID	Age	Marks	Passed
0	True	False	False	1	19.0	85.0	1
1	False	True	False	2	20.0	72.0	1
2	True	False	False	3	18.0	65.0	0
3	False	False	True	4	19.0	78.0	1
4	False	True	False	5	21.0	75.0	1
5	True	False	False	6	20.0	55.0	0
6	False	False	True	7	19.0	68.0	0
7	True	False	False	8	19.0	91.0	1
8	True	False	False	9	20.0	74.0	0
9	False	True	False	10	18.0	88.0	1

In [ ]: